

BM4322 Genomic Signal Processing
Semester 7 – 2022 Batch – 2026/2027 Academic Year
Assignment 1 – Individual - 50% of Final Grade

Promoter Discovery in Bacteria

Salmonella enterica is a rod-shaped, flagellate, facultative anaerobic, Gram-negative specie of bacteria commonly found in the gut of poultry. It causes Salmonellosis in humans. For the assignment the GenBank genome (.fasta) and its protein table (.tsv) of the given accession are required. Obtain 100 bases upstream and 3 bases downstream for 1000 genes of the sense strand. Initially for a few known genes use the presence of Methionine at the beginning of each coding sequence to verify your code. For the obtained sequence perform the following operations.

1. Perform a standard local search (for an intact query) to locate the WWW promoter within each sequence. Obtain the percentage of genes with potential promoters and the distribution of the upstream position.
2. If a WWW promoter is found find the number of consecutive Ws. Obtain the distribution of the number consecutive Ws.
3. Obtain the sequence starting from each upstream position of (1) which is 6 bases long (regardless of its C/G content). Calculate the position probability matrix.
4. Using the position probability matrix of (3) obtain a statistical alignment of the same sequences.
5. Compare the percentage of promoters detected and position distribution results of the two searches. Comment on the results.

Deliverables:

1. Brief report (.pdf via LMS) with results and discussion of the five questions.
2. Soft copies of the .fasta, .tsv and any derivative sequences (.txt or .fasta) and code have to accompany the LMS submission in a separate .zip file

Number	Student ID	Accession
1	220005R	ASM1602849v1
2	220054N	ASM155835v2
3	220074B	ASM636533v1
4	220472T	ASM2286996v1
5	220533H	ASM4108089v1
6	220563A	ASM1433415v1
7	220663F	ASM74305v1
8	220735E	ASM1556573v1

Due on: 2026.10.02